

ECSE-4040

Digital Electronics

Lecture 12

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Lecture Outline

- R_{eff} (R_{ON}) Calculation
- Capacitance Breakdown

Effective Resistance

- Terms:
 - R_{ON}
 - Resistance between S and D of single transistor
 - Same as R_{DS}
 - Typically refers to linear resistance
 - R_{eq}
 - NMOS or PMOS equivalent resistance
 - R_{eff}
 - Effective resistance of a circuit, often full pull-up, or pull-down
 - Can be one transistor, or several in series or parallel

Effective Resistance

- We have so far been using the following R_{eff} estimations for the NMOS and PMOS:

$$R_{eqn} \approx 12.5 \text{ k}\Omega / (W_N/L_N)$$

$$R_{eqp} \approx 30 \text{ k}\Omega / (W_P/L_P)$$

- When comparing k_n/k_p :
 - Long channel: $\mu_N/\mu_P = 270/70 = 3.86$
 - Short channel: $(1/E_C L_N)/(1/E_C L_P) = 4$
- $30\text{k}/12.5\text{k} = 2.4$
 - Not a simple k_n/k_p approximation

Effective Resistance

$$V_{\text{out}}(t) = V_{DD} e^{-t/RC}$$

$$\tau_{PHL} = 0.69 R_N C_L$$

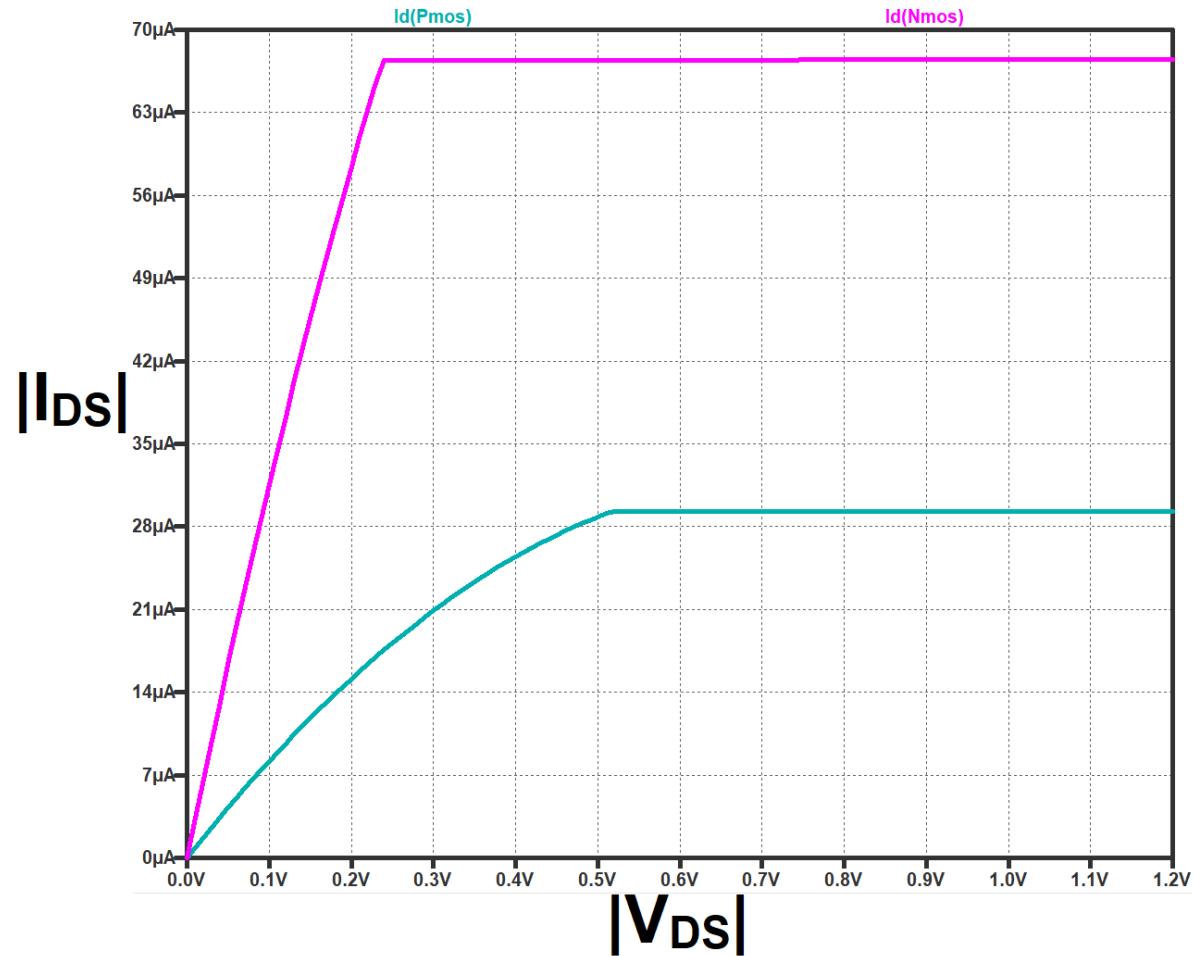
$$\tau_{PLH} = 0.69 R_P C_L$$

$$R_N = V_{DS} / I_{DS}$$

$$R_P = V_{SD} / I_{SD}$$

- We are solving this equation for when $V_{\text{OUT}}(t) = V_{DD}/2$
- Which yields 0.69 (0.7 rounded) RC
- NMOS: Pull V_{DS} (V_{OUT}) from $V_{DD} \rightarrow V_{DD}/2$ (1.2V $\rightarrow$ 0.6V)
- PMOS: Pull V_{SD} ($V_{DD} - V_{\text{OUT}}$) from $V_{DD} \rightarrow V_{DD}/2$ (1.2V $\rightarrow$ 0.6V)

Effective Resistance



Effective Resistance

$$V_{DD} = 1.2 \text{ V}, V_T = 0.4 \text{ V}, \text{ and } E_C L = 0.6 \text{ V}$$

$$V_{Dsat} = \frac{(V_{GS} - V_T)E_C L}{(V_{GS} - V_T) + E_C L} = \frac{(1.2 - 0.4)(0.6)}{(1.2 - 0.4) + (0.6)} \cong 0.34 \text{ V}$$

$$V_T = -0.4 \text{ V}, \text{ and } E_C L = 2.4 \text{ V}$$

$$V_{Dsat} = \frac{(1.2 - 0.4)2.4}{(1.2 - 0.4) + 2.4} = 0.6 \text{ V}$$

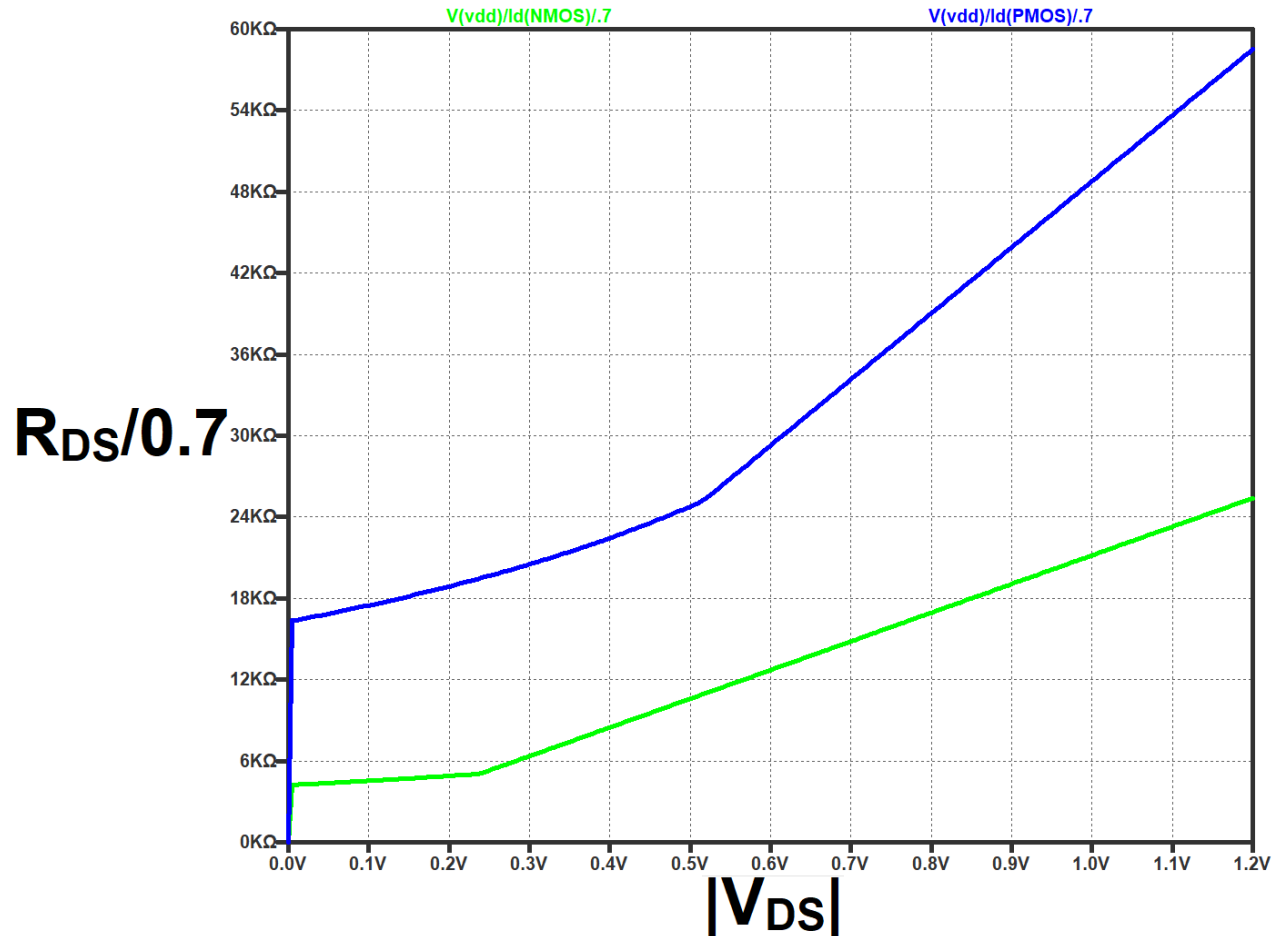
$$\tau_P = \frac{Q}{I} = \frac{C_L * \left(\frac{V_{DD}}{2}\right)}{|I_{DS}|(sat)}$$

$$R = \frac{\left(\frac{V_{DD}}{2}\right)}{0.7|I_{DS}|(sat)}$$

- For 50% output change, $|V_{DS}|$ is always $\geq 0.6 \text{ V}$
- This means saturation, and constant current

- For $R_N = 12.5 \text{ k}$ and $R_P = 30 \text{ k}$

Effective Resistance



Effective Resistance

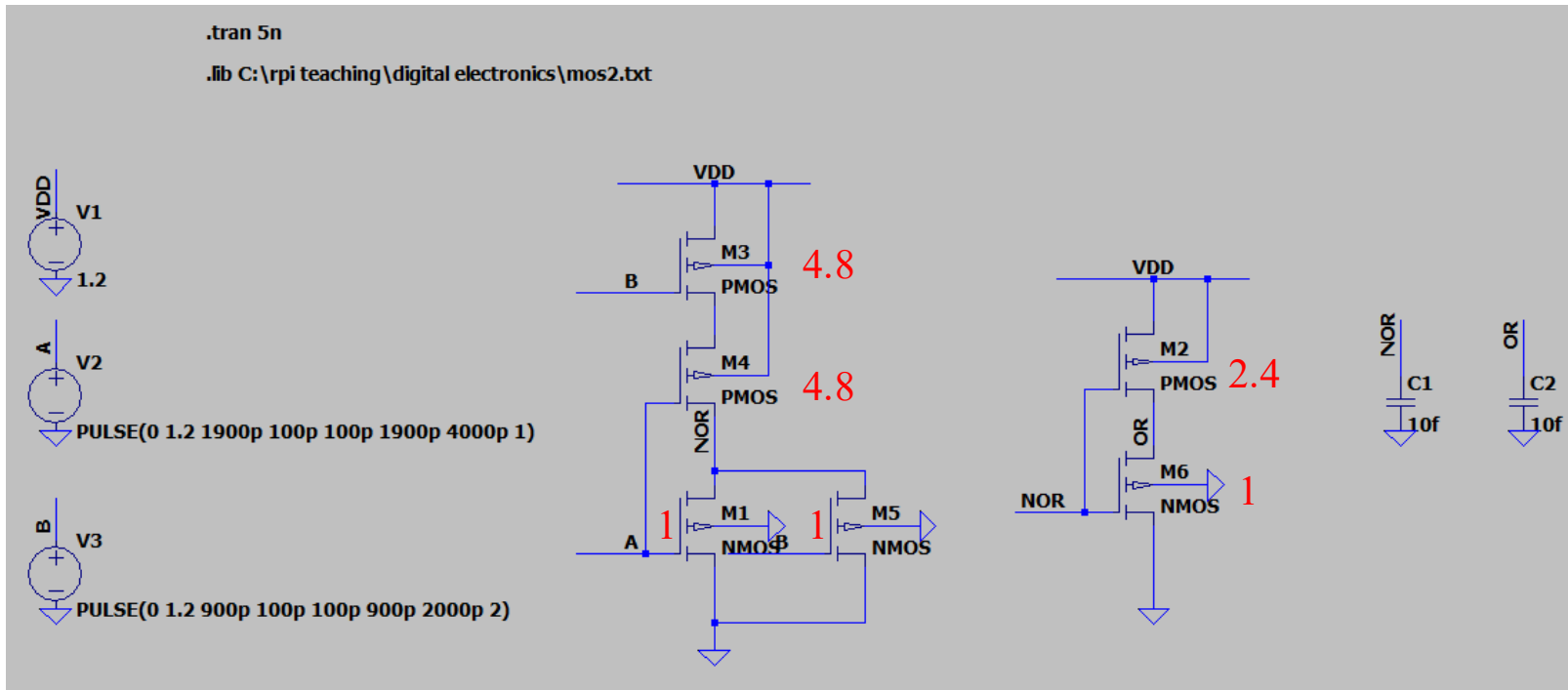
$$I_{Dsat} = \frac{W_N \mu_{sat} C_{ox} (V_{DD} - V_{TN})^2}{(V_{DD} - V_{TN}) + E_{CN} L_N}$$
$$= \frac{(0.1)(10^{-4})8(10^6)1.6(10^{-6})(1.2 - 0.4)^2}{(1.2 - 0.4) + 0.6} \approx 60 \mu A$$

$$\therefore R_{eqn} = \frac{1.2/2}{0.7(60 \mu A)} = 14.5 k\Omega$$

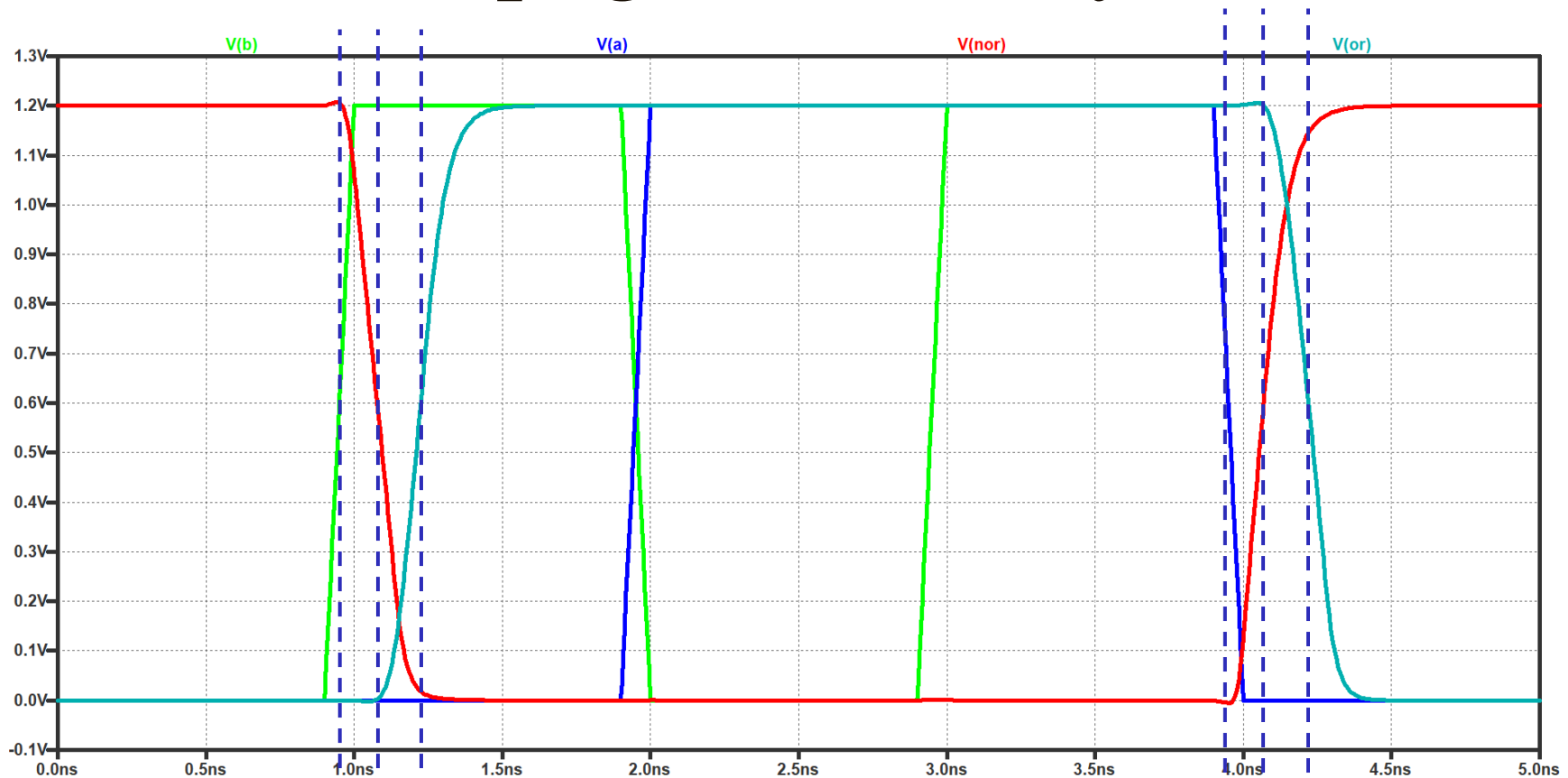
$$I_{Dsat} = \frac{W_P \mu_{sat} C_{ox} (V_{DD} - |V_{TP}|)^2}{(V_{DD} - |V_{TP}|) + E_{CP} L_P}$$
$$= \frac{(0.1)(10^{-4})8(10^6)1.6(10^{-6})(1.2 - 0.4)^2}{(1.2 - 0.4) + 2.4} \approx 25 \mu A$$

$$\therefore R_{eqp} = \frac{1.2/2}{0.7(25 \mu A)} = 33.5 k\Omega$$

Propagation Delay



Propagation Delay



$$\tau_p = 0.7RC$$

$$\tau_p = 12.5k * 10f$$

$$\tau_p = 125ps$$

1 2

$$1 = \tau_{PHL} \text{ NOR}$$

$$2 = \tau_{PLH} \text{ INV}$$

$$1+2 = \tau_{PLH} \text{ OR}$$

$$3 = \tau_{PLH} \text{ NOR}$$

$$4 = \tau_{PHL} \text{ INV}$$

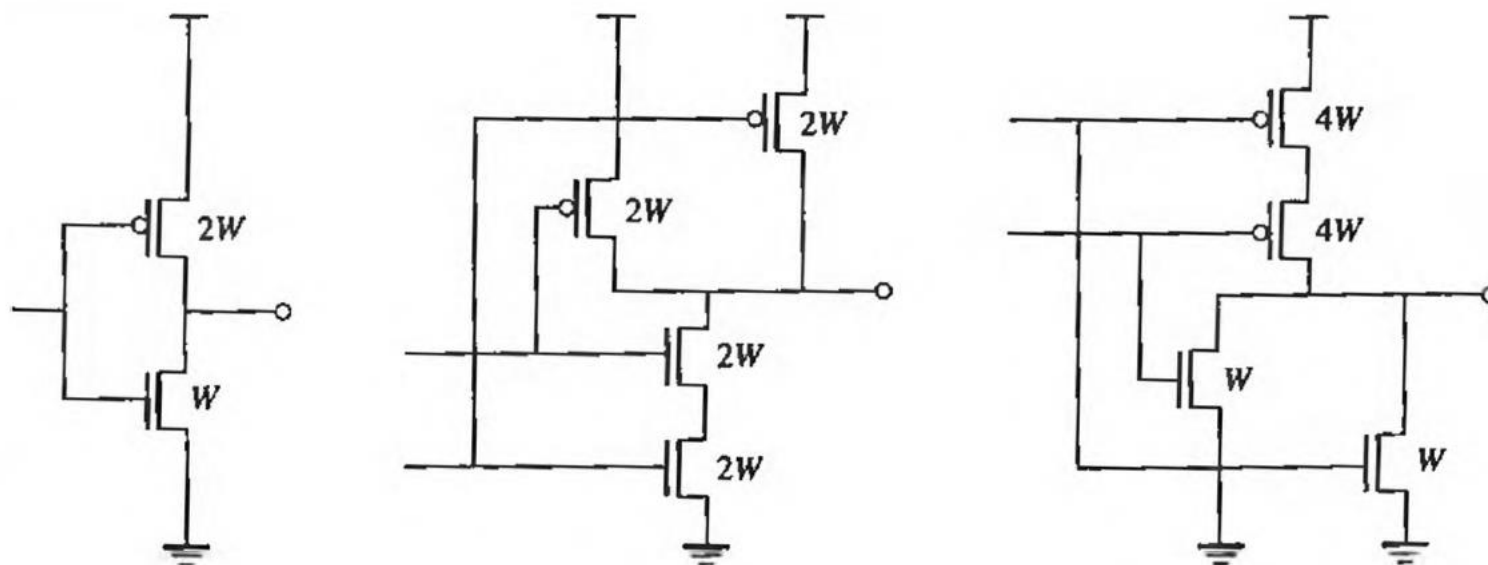
$$3+4 = \tau_{PHL} \text{ OR}$$

3 4

RPI

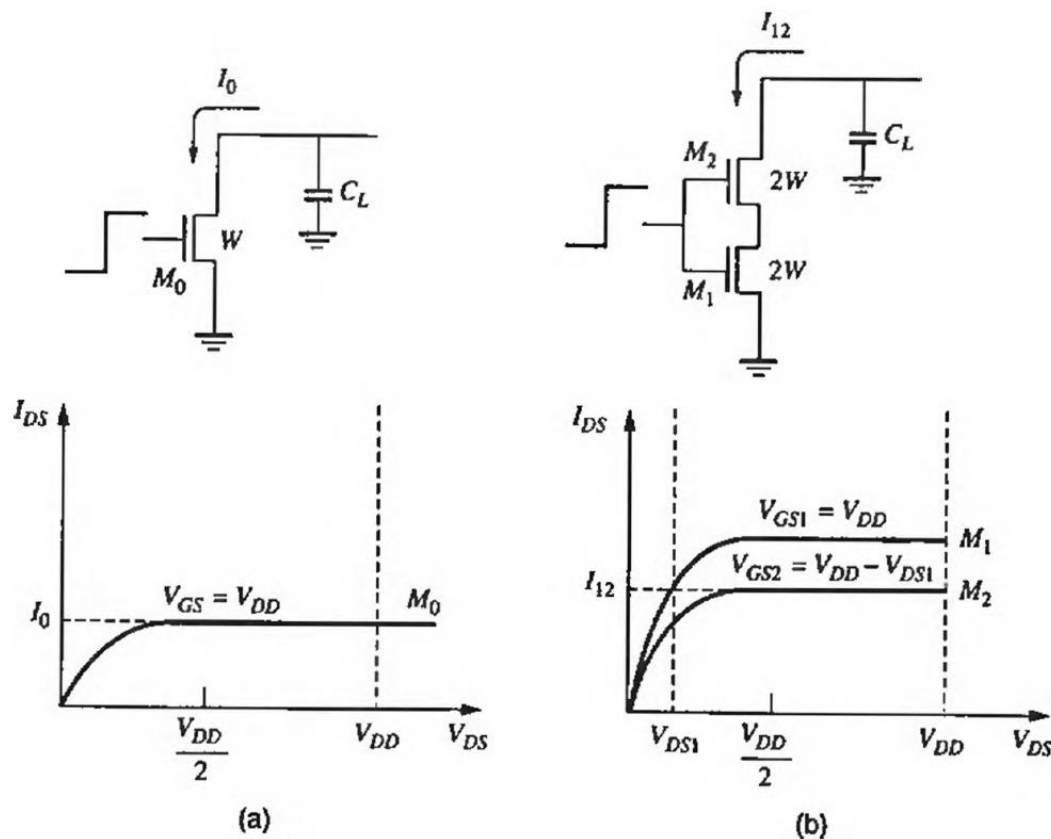
CMOS Gate Sizing Revisited

Without velocity saturation effects



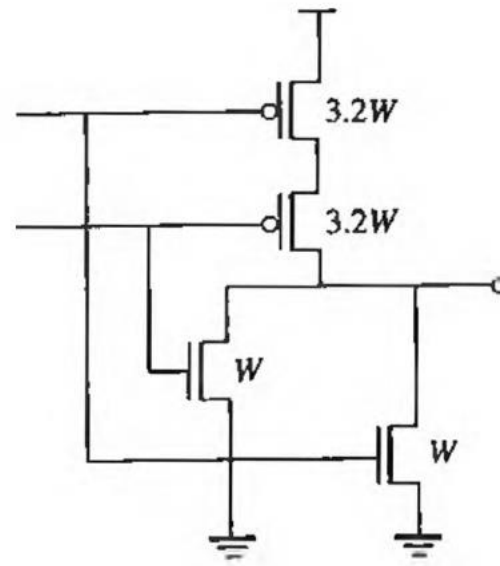
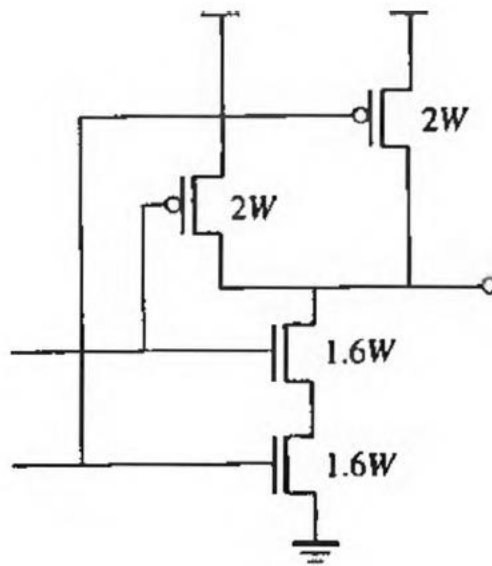
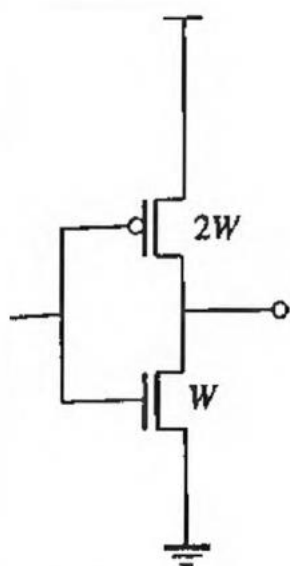
CMOS Gate Sizing Revisited

Stacked devices with velocity saturation



CMOS Gate Sizing Revisited

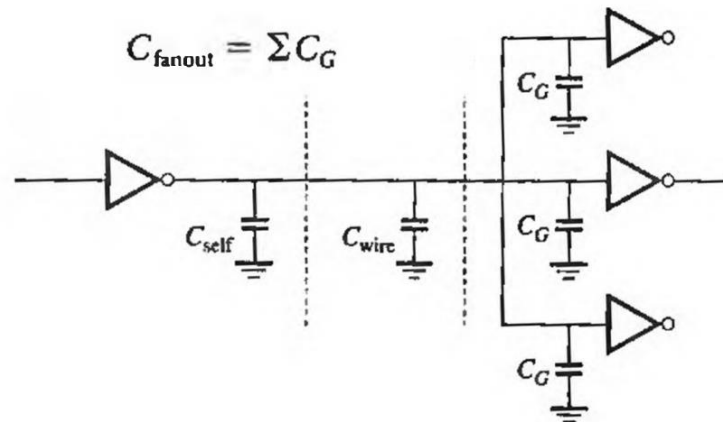
With velocity saturation effects



Load Capacitance Breakdown

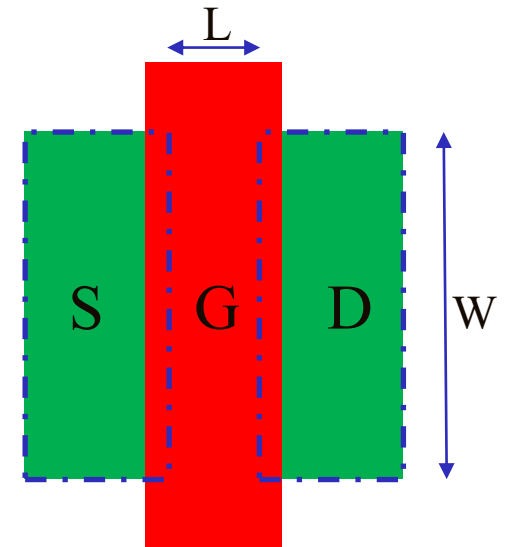
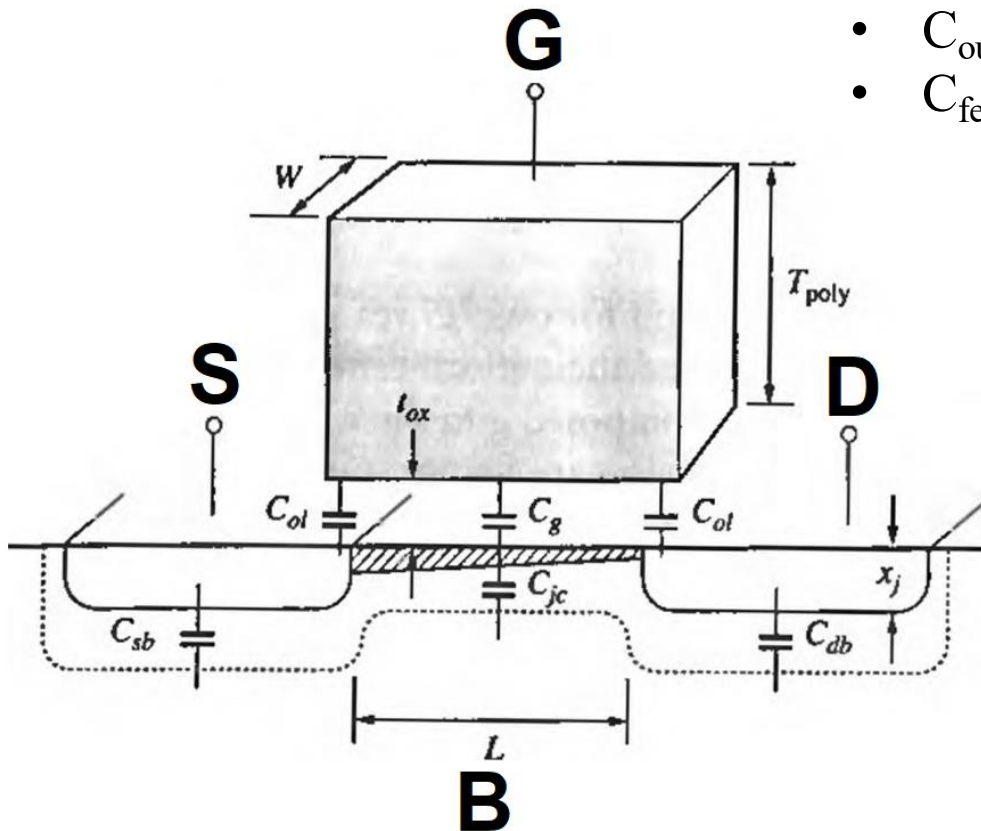
- For easy math, we have been using a single large C_L value to capture all capacitance.
- C_L has many components
 - $C_{\text{self}} = \Sigma$ Transistor output capacitance (C_{DB} dominates)
 - $C_{\text{wire}} =$ Interconnect capacitance (Metal wires = Parallel plate capacitor)
 - $C_{\text{fanout}} = \Sigma$ Input capacitance of downwind logic gates (C_{GS} dominates)

$$C_{\text{load}} = C_{\text{self}} + C_{\text{wire}} + C_{\text{fanout}}$$

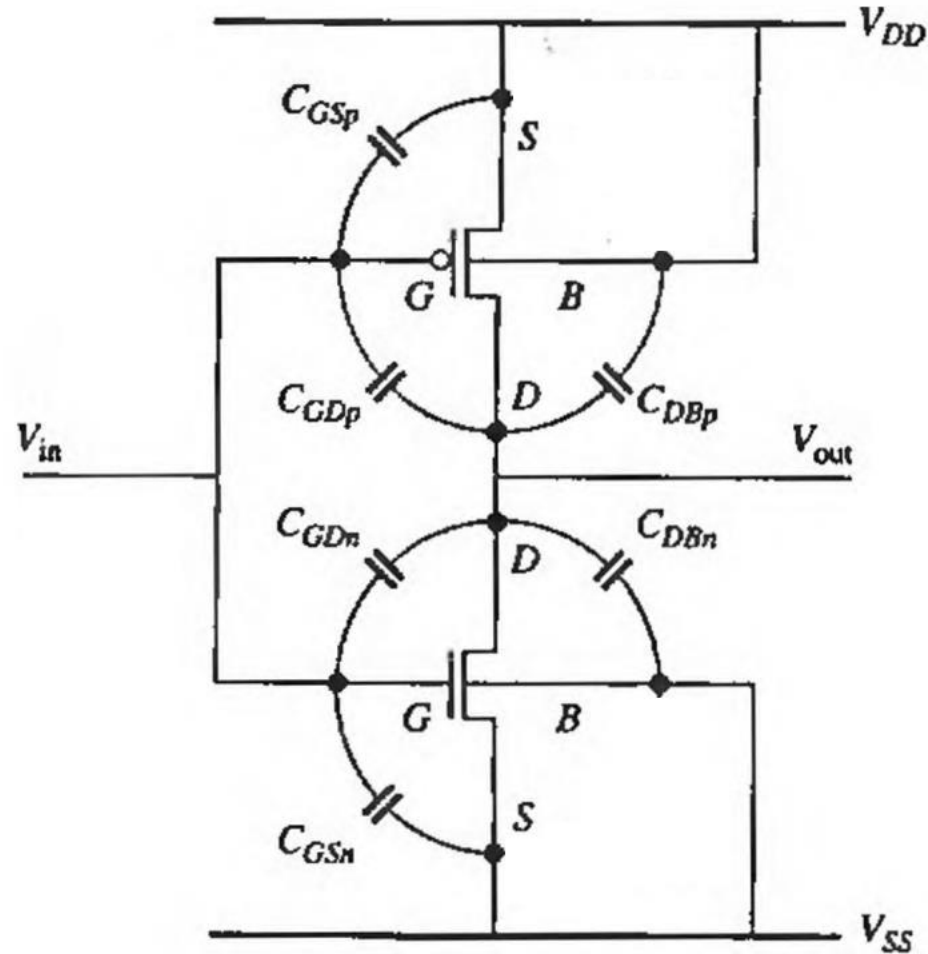
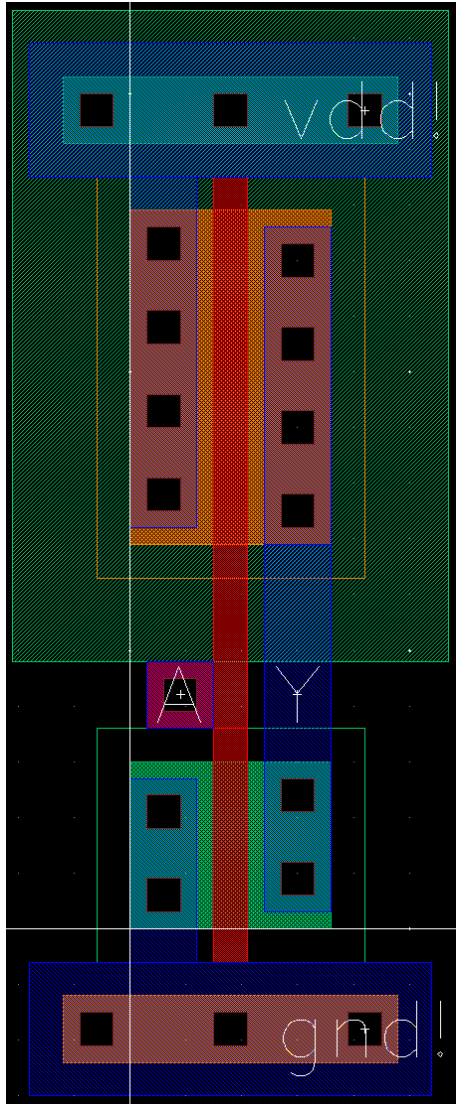


Transistor Capacitance

- $C_{in} = C_{GS} = C_G + C_{OL}$
 - $C_{OL} \sim 10\text{-}20\%$ of C_G
 - C_G is $W \cdot L \cdot C_{OX}$
- $C_{out} = W \cdot C_{DB}$
- $C_{feedthrough} = W \cdot C_{GD} = W \cdot C_{OL}$



Inverter Capacitance



Input Capacitance

- Capacitance per unit Width for a transistor:

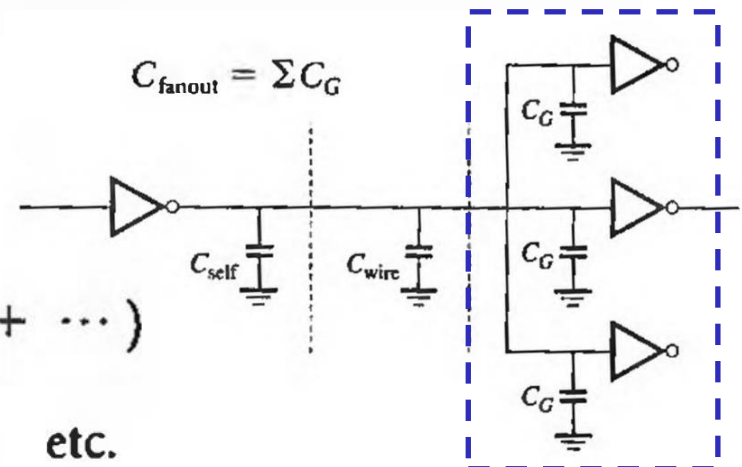
$$C_g = C_{ox}L + 2C_{ol} = 1.6 \text{ fF}/\mu\text{m} + 2(0.25 \text{ fF}/\mu\text{m}) \cong 2 \text{ fF}/\mu\text{m}$$

- Capacitance for an inverter:

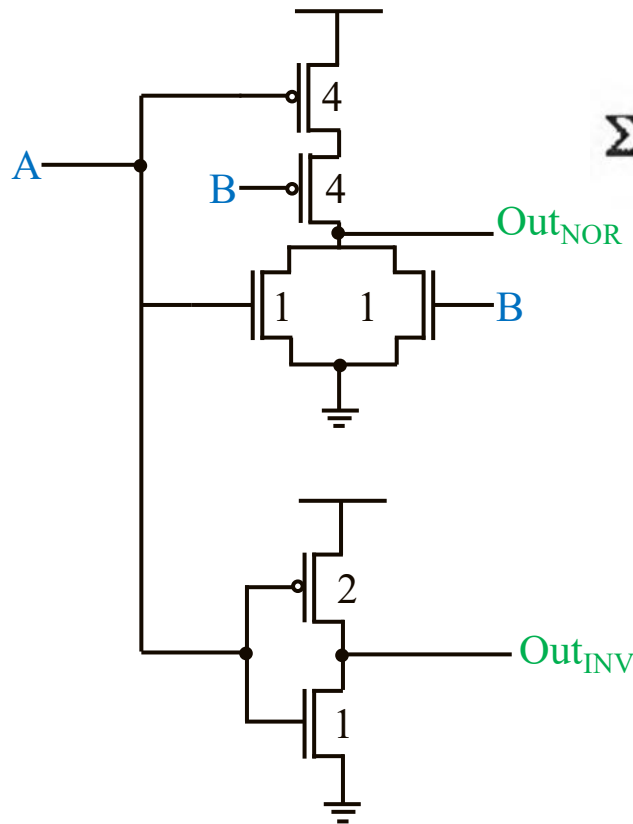
$$C_G = C_g W = 2 \text{ fF}/\mu\text{m} \times W_n + 2 \text{ fF}/\mu\text{m} \times W_p = 2 \text{ fF}/\mu\text{m} (W_n + W_p)$$

- Capacitance for complex gates:

$$\Sigma C_G = 2 \frac{\text{fF}}{\mu\text{m}} \times \underbrace{(W_{p1} + W_{n1})}_{\text{first gate}} + \underbrace{(W_{p2} + W_{n2})}_{\text{second gate}} + \dots \text{ etc.}$$



Input Capacitance



$$\Sigma C_G = 2 \frac{\text{fF}}{\mu\text{m}} \times (\underbrace{W_{p1} + W_{n1}}_{\text{first gate}} + \underbrace{W_{p2} + W_{n2}}_{\text{second gate}} + \dots)$$

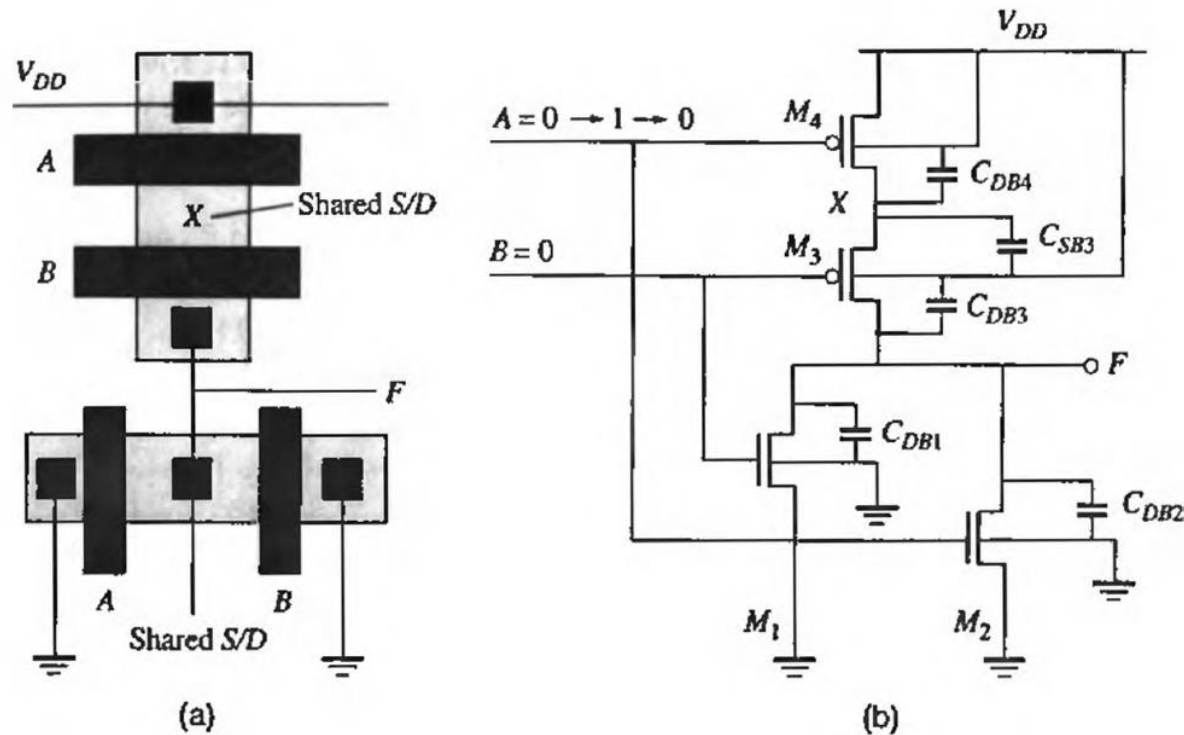
- Input capacitance seen at A and B:

W = 100nm at minimum. With $2 \frac{\text{fF}}{\mu\text{m}}$, we have 0.2fF/100nm

$$\Sigma C_{GB} = 0.2\text{f} \cdot (4 + 1 + 2 + 1) = 1.6\text{fF}$$

$$\Sigma C_{GB} = 0.2\text{f} \cdot (4 + 1) = 1\text{fF}$$

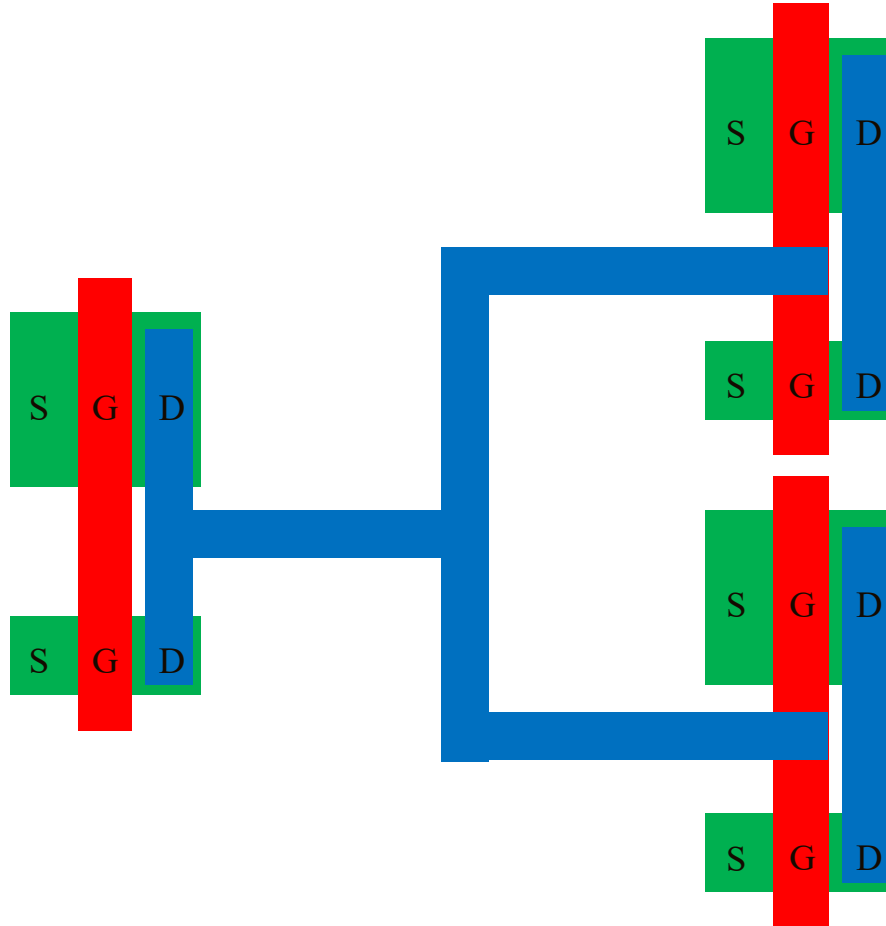
Self Capacitances for NOR Gate



$$C_{\text{self}} = \underbrace{C_{DB1} + C_{DB2}}_{n^+ \text{ shared S/D}} + \underbrace{C_{DB3} + C_{SB3} + C_{DB4}}_{p^+ \text{ shared S/D}}$$

$$= C_{DB12} + C_{DB3} + C_{SDB34}$$

Wire Capacitance



- C_{wire} is blue
- Parallel plate capacitor to GND

$$C_{\text{wire}} = C_{\text{int}} L_W = 0.2 \text{ fF}/\mu\text{m} \times (\text{wirelength})$$

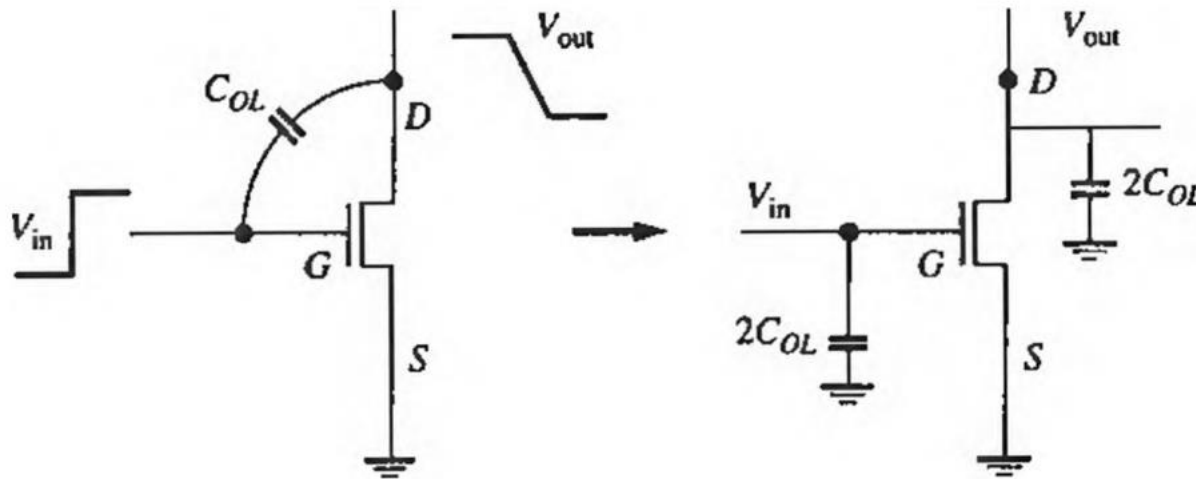
For 10 μm wire:

$$C_{\text{wire}} = 0.2 \text{ f} \cdot 10 = 2 \text{ fF}$$

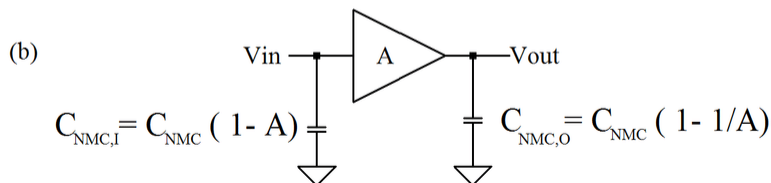
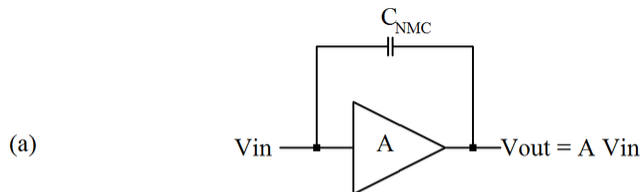
For 1 mm wire:

$$C_{\text{wire}} = 0.2 \text{ f} \cdot 1000 = 200 \text{ fF}$$

Feedthrough Capacitance



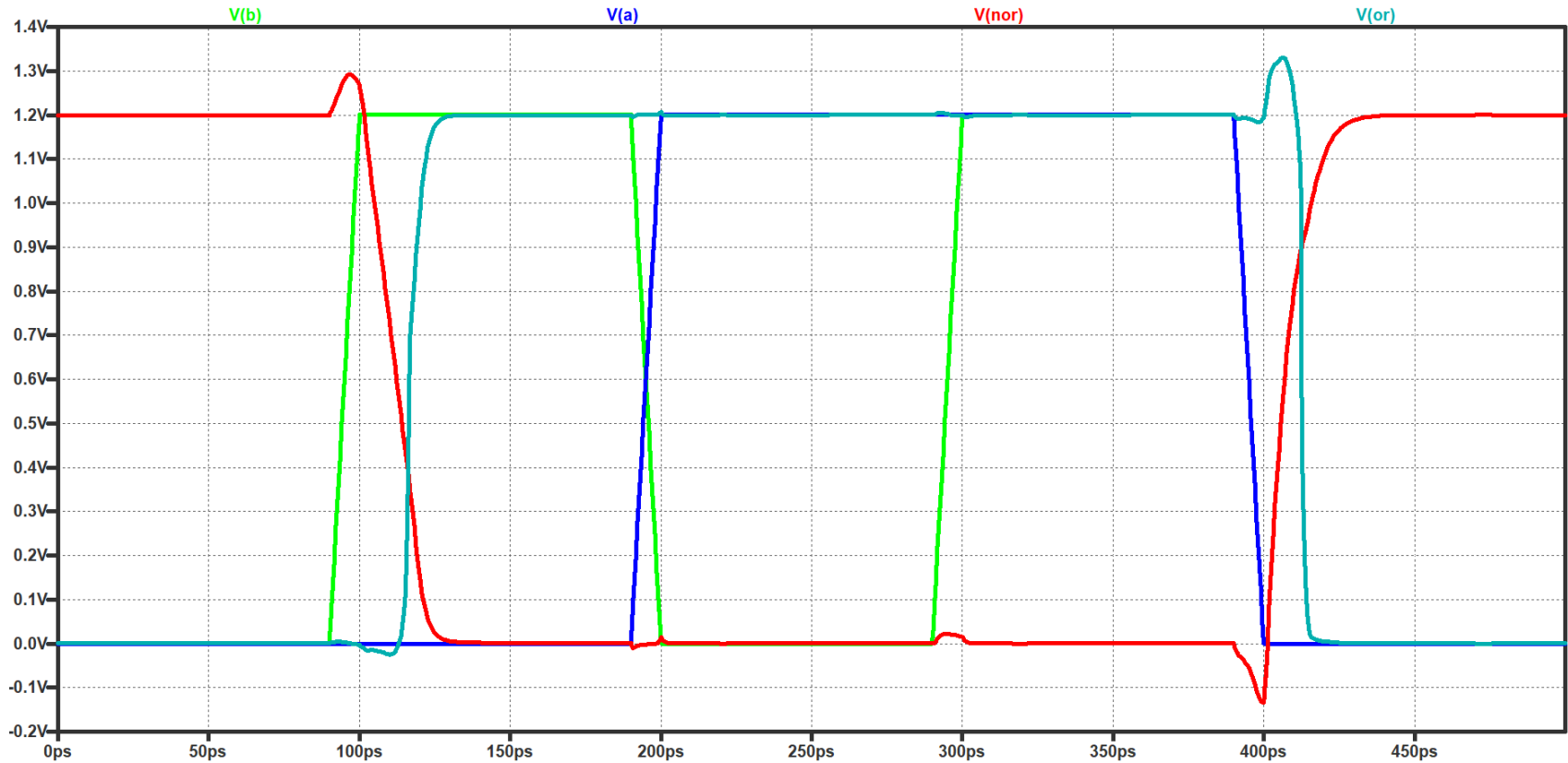
- Feedthrough/overlap capacitance causes effective input and output capacitance increase.
 - Substantial for output
- Modelled with Miller effect
- Ignores feedthrough effect



Here $A = -1$

https://www.researchgate.net/figure/a-Non-inverting-amplifier-with-negative-Miller-capacitor-and-b-Negative-Miller_fig3_326155046

Feedthrough Capacitance



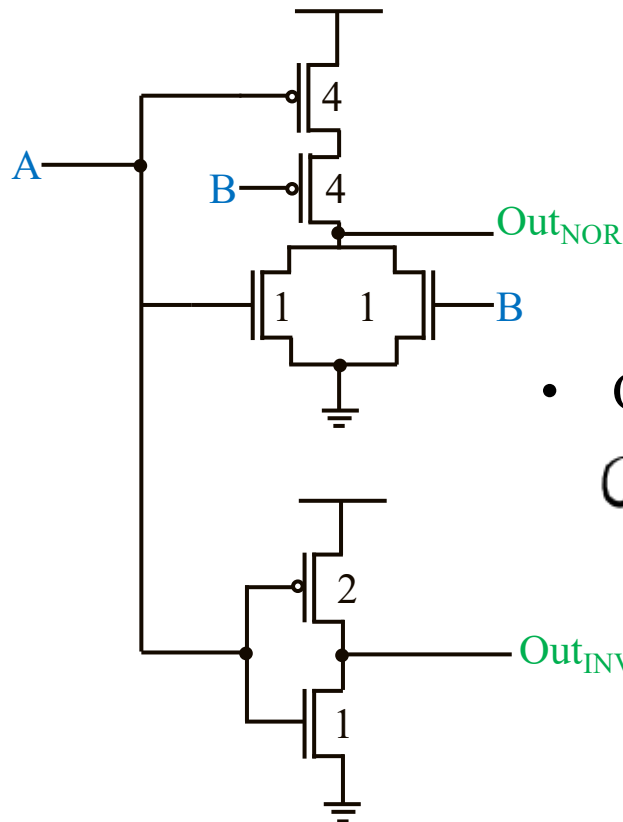
Output Capacitance

$$\begin{aligned}C_{\text{self}} &= C_{DBn} + C_{DBp} + 2C_{OL} + 2C_{OL} \\&= C_{jn}W_n + C_{jp}W_p + 2C_{ol}(W_n + W_p) \\&= C_{\text{eff}}(W_n + W_p)\end{aligned}$$

$$C_{\text{eff}} = C_j + 2C_{ol} \approx 0.5 \text{ fF}/\mu\text{m} + 2(0.25 \text{ fF}/\mu\text{m}) \approx 1 \text{ fF}/\mu\text{m}$$

$$C_{\text{output}} = C_{\text{self}} = (W_n + W_p) * 1 \text{ fF}/\mu\text{m}$$

Output Capacitance



$$C_{\text{output}} = C_{\text{self}} = (W_n + W_p) * 1 \text{ fF}/\mu\text{m}$$

W = 100nm at minimum. With $2 \frac{\text{fF}}{\mu\text{m}}$, we have 0.1 fF/100nm

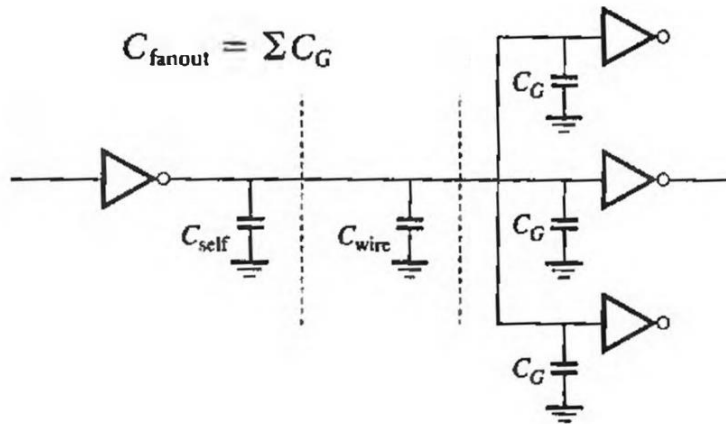
- Output capacitance seen at OUT_{NOR} and OUT_{INV} :

$$C_{\text{NOR}} = \underbrace{[(1+1) + 4]}_{\text{1 in layout}} \cdot 0.1\text{f} = 0.6\text{fF}$$

$$C_{\text{NOR(int)}} = 4 \cdot 0.1\text{f} = 0.4\text{fF} \leftarrow \text{Node between PMOS}$$

$$C_{\text{INV}} = (1+2) \cdot 0.1\text{fF} = 0.3\text{fF}$$

Inverter Fanout Example



Final Load Capacitance:

$$C_{load} = C_{self} + C_{wire} + C_{fanout}$$

$$C_{load} = 0.3 \text{ fF} + 2 \text{ fF} + 1.8 \text{ fF}$$

$$C_{load} = 4.1 \text{ fF}$$

Capacitance from transistor output:

$$C_{self} = C_{output} = C_{self} = (C_j + 2C_{ol}) * (W_n + W_p) \\ = 1 \text{ fF}/\mu\text{m} * (W_n + W_p)$$

$$C_{self} = C_{out} = (1+2) \cdot 0.1 \text{ fF} = 0.3 \text{ fF}$$

Capacitance from transistor input:

$$\Sigma C_G = 2 \frac{\text{fF}}{\mu\text{m}} \times (\underbrace{W_{p1} + W_{n1}}_{\text{first gate}} + \underbrace{W_{p2} + W_{n2}}_{\text{second gate}} + \dots \text{etc.})$$

$$C_{in} = \Sigma C_G = 0.2 \text{ fF} \cdot (2+1 + 2+1 + 2+1)$$

$$C_{fanout} = C_{in} = 1.8 \text{ fF}$$

Capacitance from wiring (10 μm):

$$C_{wire} = C_{int} L_W = 0.2 \text{ fF}/\mu\text{m} \times (\text{wirelength})$$

$$C_{wire} = 0.2 \text{ fF} \cdot 10 = 2 \text{ fF}$$

NOR, OR C_L Example

$$W_{M3} = W_{M4} = 4.8$$

$$W_{M2} = 2.4$$

$$W_{M1} = W_{M5} = W_{M6} = 1$$

NOR node:

Capacitance from transistor output:

$$C_{self} = C_{output} = C_{self} = (W_n + W_p) \cdot 1 \text{ fF}/\mu\text{m} \\ = (1 + 4.8) \cdot 0.1 \text{ f} = 0.68 \text{ fF}$$

Capacitance from transistor input:

$$\Sigma C_G = 2 \frac{\text{fF}}{\mu\text{m}} \times (\underbrace{W_{p1} + W_{n1}}_{\text{first gate}} + \underbrace{W_{p2} + W_{n2}}_{\text{second gate}} + \dots)$$

$$\Sigma C_G = 0.2 \text{ f} \cdot (2.4 + 1) = 0.68 \text{ fF}$$

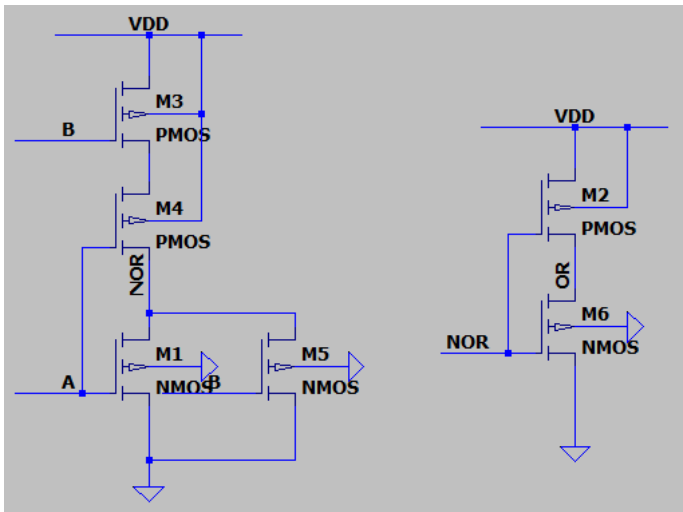
$$C_{NOR} = C_{self} + \Sigma C_G$$

$$C_{NOR} = 1.36 \text{ fF}$$

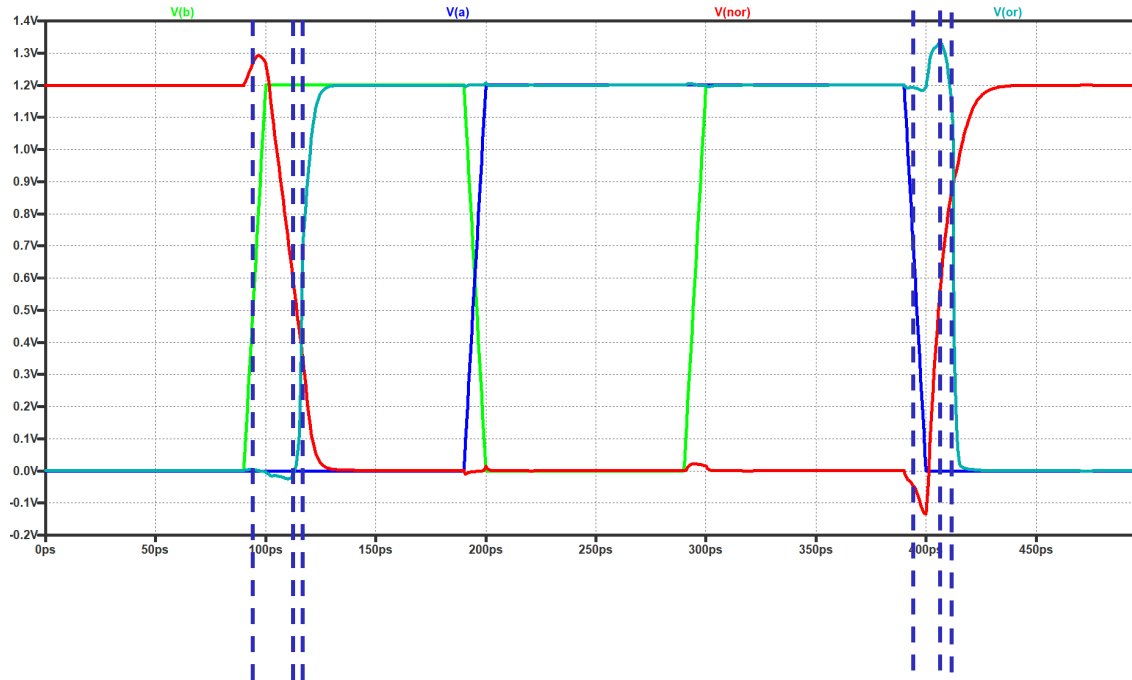
OR node (Inverter):

$$C_{self} = (1 + 2.4) \cdot 0.1 \text{ f} = 0.34 \text{ fF}$$

$$C_{OR} = 0.34 \text{ fF}$$



NOR, OR Delay



- LTSPICE Capacitance parameters has not been configured to match HJS values
- Overall ratios match

$$\frac{170}{42.5} = 4$$

$$\begin{aligned}\tau_{PNOR} &= 0.7RC \\ \tau_{PNOR} &= 12.5k * 13.6f \\ \tau_{PNOR} &= 170 \text{ ps}\end{aligned}$$

$$\begin{aligned}\tau_{PINV} &= 0.7RC \\ \tau_{PINV} &= 12.5k * 3.4f \\ \tau_{PINV} &= 42.5 \text{ ps}\end{aligned}$$

$$\begin{aligned}\tau_{POR} &= \tau_{PNOR} + \tau_{PINV} \\ \tau_{POR} &= 170p + 42.5p \\ \tau_{POR} &= 212.5 \text{ ps}\end{aligned}$$